

Thesis Synopsis

Inference-Time Coordination of Foundation-Model Representations for Robust Visual Perception under Limited Supervision

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Research Problem

Although foundation models have significantly advanced visual perception, adaptation to downstream tasks remains predominantly dependent upon parameter optimisation through fine-tuning, prompt learning, adapters, or meta-learning. These approaches increase computational cost, require labelled data, and reduce deployment flexibility in low-supervision environments.

This research investigates whether pretrained foundation-model representations can instead be adapted through inference-time coordination without modifying model parameters.

Research Hypothesis

Robust visual perception under limited supervision can be achieved by coordinating complementary pretrained representations during inference rather than through task-specific parameter optimisation.

Research Contributions

The thesis develops a unified computational framework for inference-time coordination across three complementary representation domains.

1. Feature-space coordination through Test-Time Iterative Prototype Mining (TIPM) for generalized few-shot semantic segmentation.
2. Geometric coordination through Cross-View Semantic Consensus for generalized few-shot three-dimensional point-cloud segmentation.
3. Semantic coordination through OpenV-GFSS, a training-free open-vocabulary generalized few-shot semantic segmentation framework integrating promptable segmentation and vision-language foundation models.

Experimental Validation

The proposed methods are evaluated on multiple benchmark datasets including PASCAL-5i, COCO-20i, ADE-based open-world datasets, and ScanNet++. Experimental results demonstrate that competitive visual perception performance can be achieved through inference-time coordination without task-specific parameter optimisation.

Research Impact

The principal contribution of the thesis is the establishment of inference-time coordination as a unified paradigm for adapting pretrained foundation-model representations across feature, geometric, and semantic domains. The research provides a computationally efficient alternative to optimisation-based adaptation and establishes a foundation for future work in adaptive foundation models and multimodal reasoning.